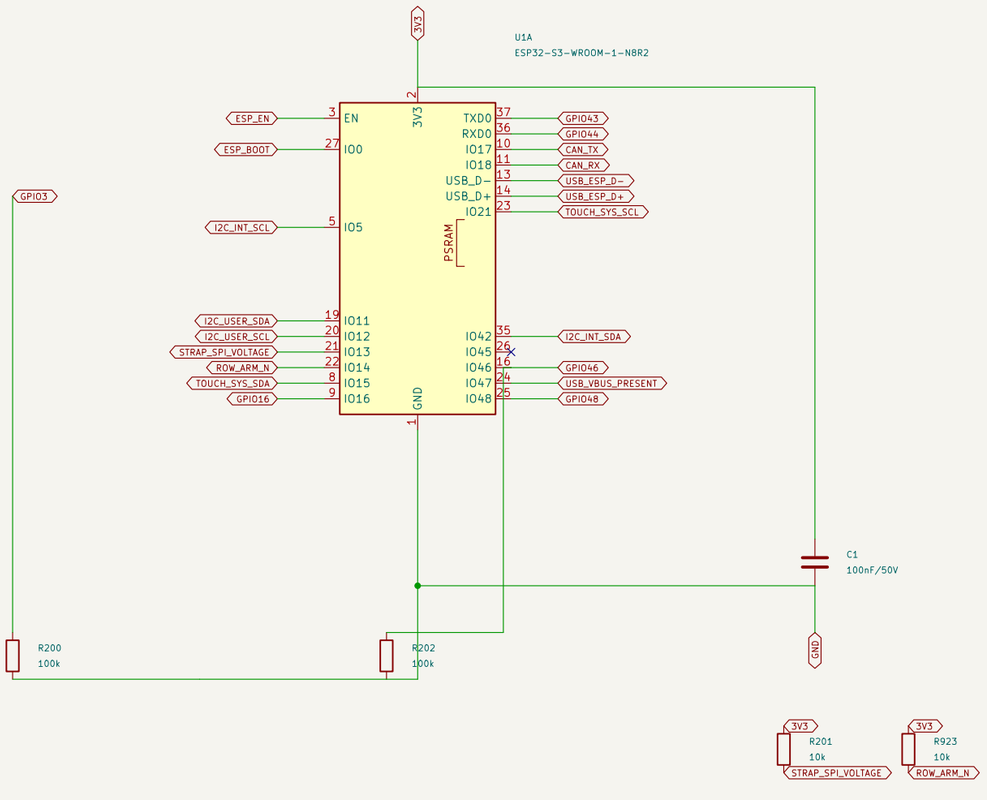
SPAGHETTI LAB | 16 RIGHE | S3-N8R2 / S31-WROOM-1

ESP32 / FLEXY PIN (symbol GPIO names = S3 aliases)

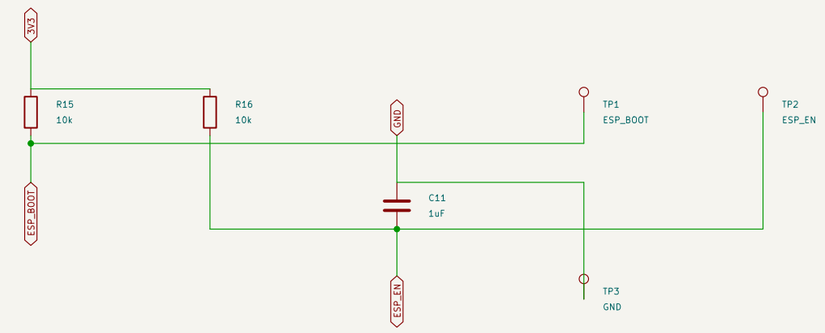
ADC: S3 A1-A5/B1-B4; S31 A1/A6-A8/B1/B5-B8.
U1 row units B..Q are wired on PORTS16. Internal SDA: pad35.



DUAL MCU: R201 fixes S31 GPIO36 high. Pad26 has NO external pull.
R923 holds ROW_ARM_N high: rows OFF during boot/programming.
J25 pins7/8/12 reserved NC; never export flash strap or local touch bus.

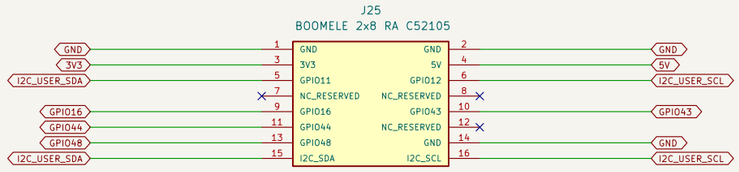
BOOT, EN E PIAZZOLE DI SERVIZIO

BOOT pad27: S3 GPIO0 / S31 GPIO61. ROW_ARM_N is separate: see control sheet.



ESPANSIONE SUD / BOOMELE C52105 2x8

3.3V direct GPIO: names GPIOxx are S3 aliases. J25 7/8/12 NC. User I2C only.



Pin1 quadrato = GND. Passo e distanza tra file: 2.54mm.

5V solo alimentazione; GPIO non tolleranti 5V. Nessuna commutazione delle righe.

LEGENDA COLLEGAMENTI GLOBALI
3V3: rail regolata; 5V: rail ingresso; GND: riferimento comune.
I2C_INT: pad4/5 internal devices; I2C_USER: pad19/20 connectors; CAN_TX/RX: pad10/11.
USB_ESP_D+/D-: coppia USB verso ESP; USB_VBUS_PRESENT: rilevamento5VUSB.
ROW_ARM_N: pad22, high=rows OFF; EXP_RESET_N: reset expander and row power enable.
ROW_Lxx = banco A; ROW_Rxx = banco B. Suffissi spiegati nel foglio05.
Tutti i nomi sono globali; i fili mostrano la topologia. Punto = connessione, incrocio senza punto = nessun collegamento.

02 – Alimentazione e USB
File: power-usb.kicad_sch

03 – CAN e terminazione automatica
File: can.kicad_sch

04 – Controllo righe, reset sicuro e Qwiic
File: control.kicad_sch

05 – Sedici righe programmabili
File: ports16.kicad_sch

06 – Indicatori di configurazione retroilluminati
File: rgb-ring.kicad_sch

07 – Mezzanina passiva
File: mezzanine.kicad_sch

08 – Selezione, comandi e logo
File: ui-indicators.kicad_sch

U91: 16 panel keys on dedicated software I2C. U90 removed.
See PANEL16-FFC60.md for key mapping, STOP and accessory output interlock.

10 – Global touch commands
File: touch-global.kicad_sch

11 – Blue logo,16 LEDs
File: logo16.kicad_sch

12 – Panel LEDs
File: panel-leds.kicad_sch

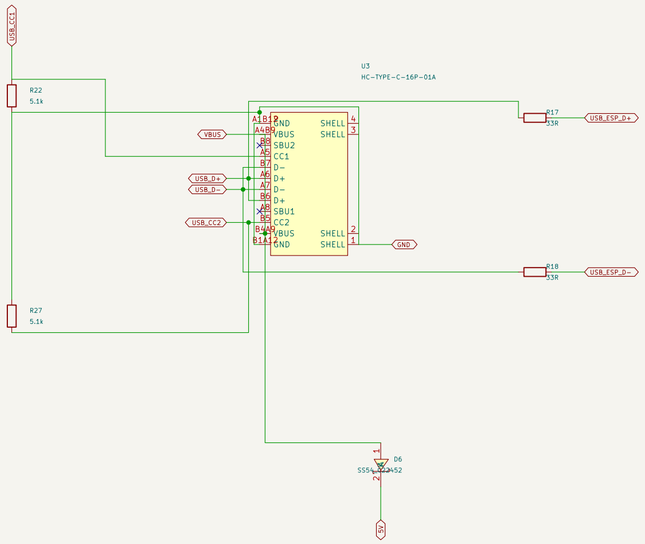
13 – FFC60 accessory
File: expansion-ffc60.kicad_sch

Sheet: /
File: backbone.kicad_sch
Title: 01 – ESP32 e boot
Size: User | Date: 2026-11-09 | **Rev: 16R-SELECTOK4**
KiCad E.D.A. 10.0.3 | Id: 1/12

USB-C E ALIMENTAZIONE 5V / 3V3

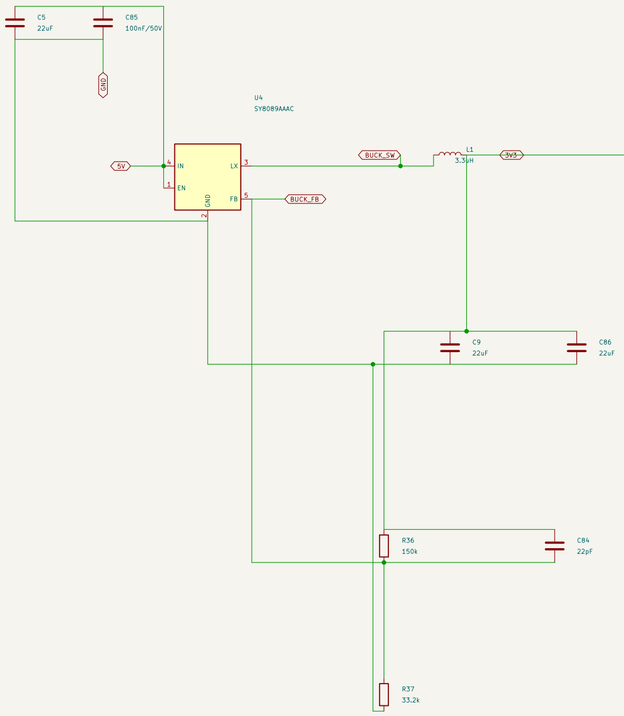
USB-C, CC E DATI

CC1/CC2: resistenze dispositivo USB; D6: alimentazione da VBUS.



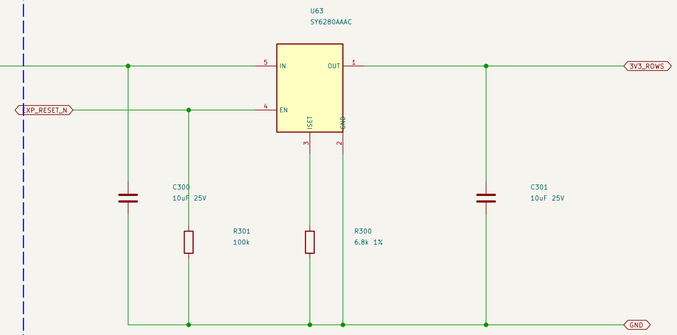
REGOLATORE 5V -> 3V3

BUCK_SW: nodo commutato; BUCK_FB: retroazione del regolatore.
Le etichette 5V/3V3/GND connettono gli altri fogli.



PROTEZIONE COMUNE ALIMENTAZIONE RIGHE

3V3_ROWS alimenta solo i rami VCC delle16righe.
ESP32, CAN, Qwiic e connettori diretti restano su3V3.

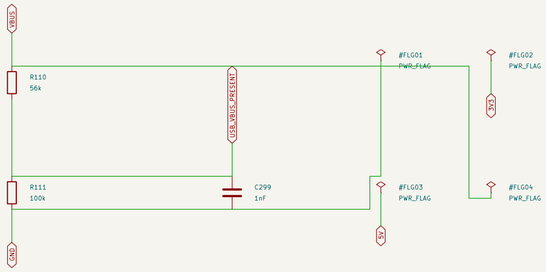


R300: ILIM=6800/R; 6.8k -> 1A nominale condiviso.
Datasheet:0.75..1.25A a5V/25C;500mA e3.3V da validare.
Corto persistente: limite corrente, arresto termico e riprova.
Non protegge GPIO,5V esterni o le rail dirette dei connettori.
Il chip non fornisce un segnale FAULT. EN non deve fluttare.

Avvio: EXP_RESET_N basso -> U63 spento.
Senza scarica attiva (versioneA): i condensatori possono restare carichi. Righe Hi-Z garantite solo dalle sequenze e dai pull del circuiti esistenti, da provare sul prototipo.
GND rimane comune. PCB ancora da sbrogliare.

RILEVAMENTO VBUS E FLAG ALIMENTAZIONI

USB_VBUS_PRESENT: partitore56k/100k verso pad24 (53 GPIO47 / 531 GPIO39). filtro1nF.

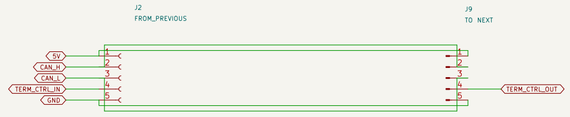


Sheet: /02 – Alimentazione e USB/		
File: power-usb.kicad_sch		
Title: 02 – Alimentazione e USB		
Size: A2	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 2/12

BUS CAN E TERMINAZIONE AUTOMATICA

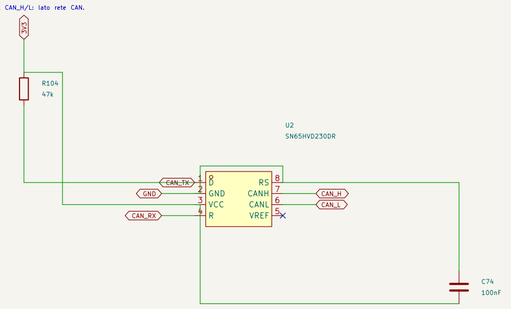
CONNETTORI MAGNETICI PASSANTI

CAN_H/L: bus differenziale. TERM_CTRL_IN/OUT: rilevamento dei vicini.



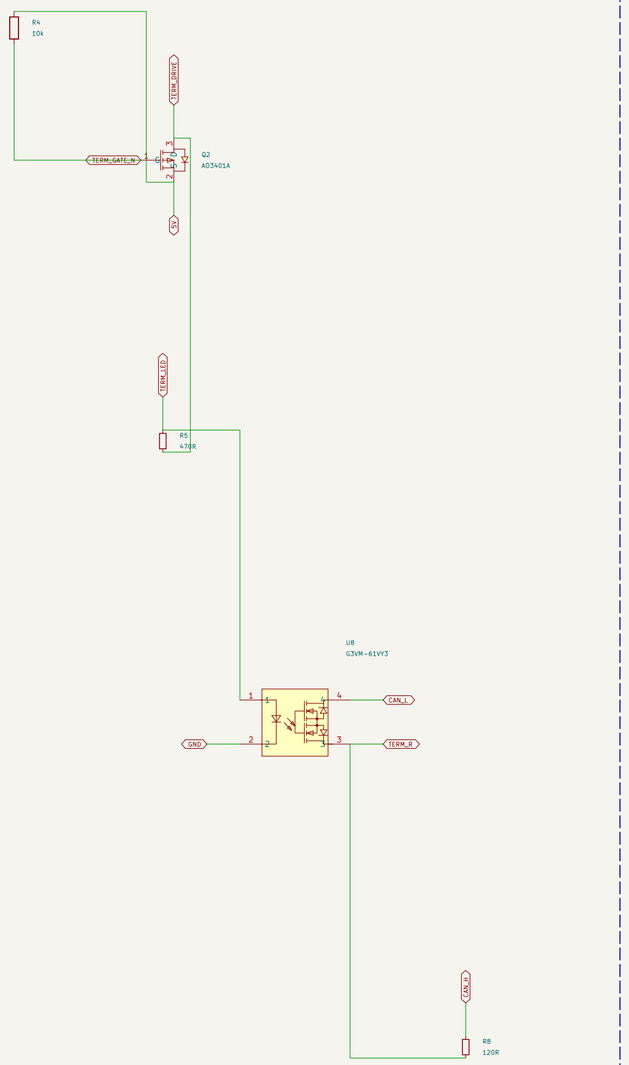
TRANSCEIVER CAN 3V3

CAN_TX/RX: interfaccia ESP: CAN_H/L: lato rete CAN.



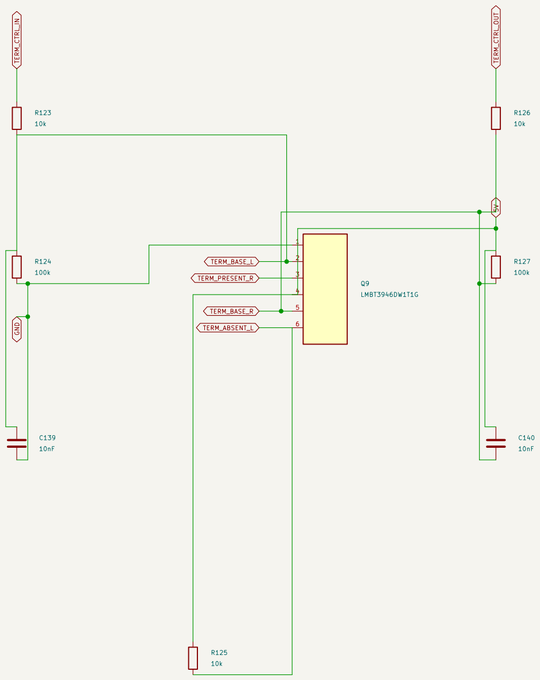
INTERRUTTORE E RESISTENZA DI TERMINAZIONE

TERM_DRIVE / TERM_LED: pilotaggio foto-MOS.
TERM_R: collegamento tra120ohm e interruttore verso CAN_L.
Quando abilitata, la resistenza120ohm e fra CAN_H e CAN_L.



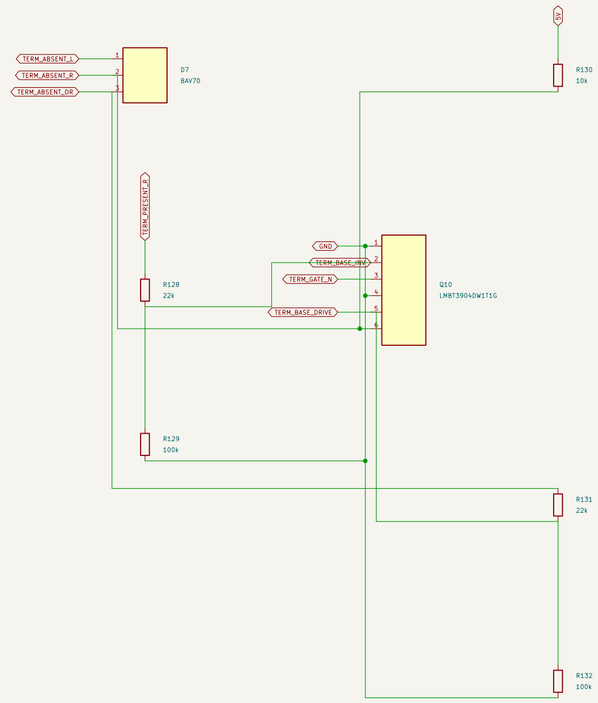
RILEVAMENTO VICINO SINISTRO / DESTRO

TERM_BASE_L/R: nodi di filtro.
TERM_ABSENT_L: assenza a sinistra; TERM_PRESENT_R: presenza a destra.

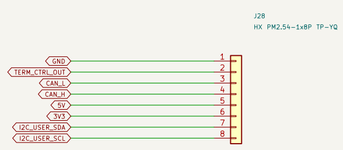


COMBINAZIONE ASSENZE E COMANDO

TERM_ABSENT_R e TERM_ABSENT_OR: estremità rilevate.
TERM_GATE_N: comando attivo basso del PMOS di terminazione.

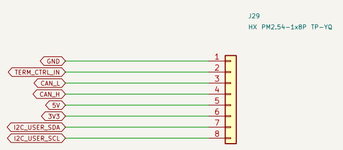


ESPANSIONE BUS J28 / MAGNETICO J9



Pin 1..5 condividono i fori e le reti del magnetico.
6:3V3 / 7:SDA / 8:SCL. I2C solo locale a3.3V.
DET mantiene la funzione originale di terminazione CAN:
non usarlo come GPIO e non forzarlo sulla espansione.
Non alimentare contemporaneamente il bus da sorgenti
non isolate. Nessuna protezione aggiunta su questi pin.

ESPANSIONE BUS J29 / MAGNETICO J2



Pin 1..5 condividono i fori e le reti del magnetico.
6:3V3 / 7:SDA / 8:SCL. I2C solo locale a3.3V.
DET mantiene la funzione originale di terminazione CAN:
non usarlo come GPIO e non forzarlo sulla espansione.
Non alimentare contemporaneamente il bus da sorgenti
non isolate. Nessuna protezione aggiunta su questi pin.

Sheet: /03 – CAN e terminazione automatica/
File: can.kicad_sch

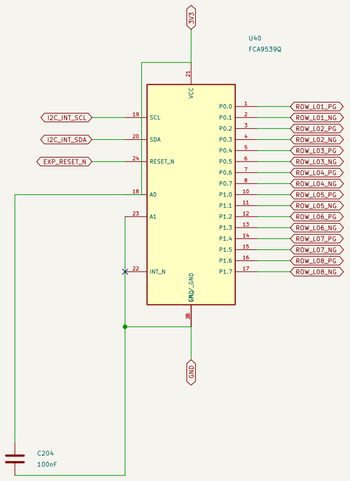
Title: 03 – CAN e terminazione automatica

Size: User	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 3/12

CONTROLLO48SEGNALI / RESET SICURO / I2C

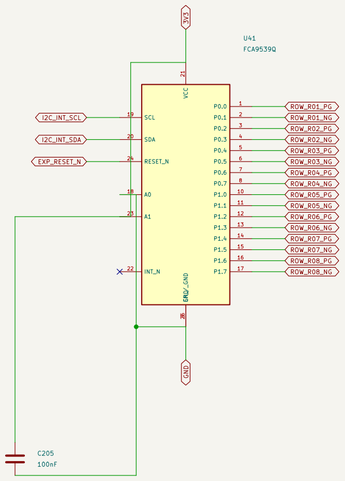
U40 | FCA9539Q | 0x75

PG/NG banco A: comandi di alimentazione.
SCL/SDA: I2C; A0/A1: indirizzo; INT_N non usato.



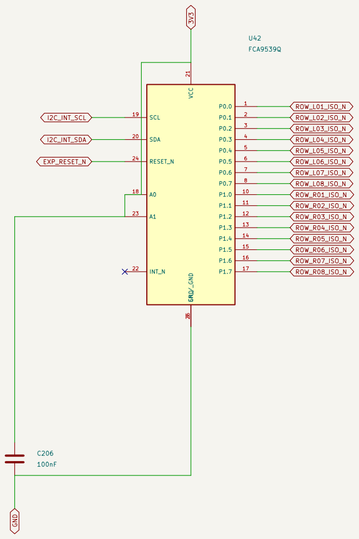
U41 | FCA9539Q | 0x76

PG/NG banco B: comandi di alimentazione.
SCL/SDA: I2C; A0/A1: indirizzo; INT_N non usato.



U42 | FCA9539Q | 0x77

ISO_N A/B: isolamento GPIO, attivo basso.
SCL/SDA: I2C; A0/A1: indirizzo; INT_N non usato.



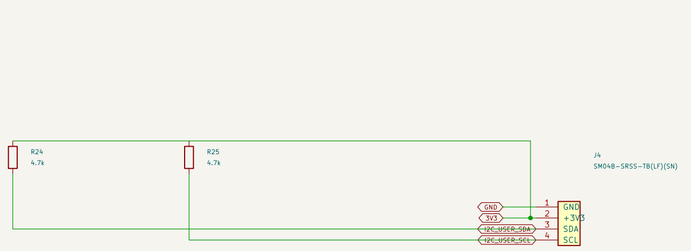
INTERBLOCCO RESET / UNA SOLA PORTA

EXP_RESET_N = ESP_EN AND NOT ROW_ARM_N. IN0 a GND.
R203 mantiene reset basso se non pilotato; C200 bypass100nF.



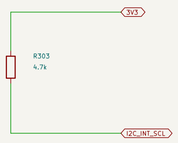
BUS UTENTE GPIO11/12 / QWIIC / PULL-UP

I2C_USER indipendente dall'expander: SDA GPIO11 / SCL GPIO12. Pull-up4.7k a3V3.
3V3/GND: alimentazione del moduloQwiic.



BUS INTERNO DEDICATO / GPIO4 SDA / GPIO5 SCL





R302/R303: pull-up4.7k del solo bus interno.
R24/R25: pull-up4.7k del bus utente, nel blocco Qwiic.
Nessun collegamento SDA/SCL tra i due bus.
Entrambi usano livelli3.3V: firmware: due controller distinti.

INTERFACCIA LOCALE | DUE PULSANTI XKB

Sheet: /04 – Controllo righe, reset sicuro e Qwiic/	
File: control.kicad_sch	
Title: 04 – Controllo righe, reset sicuro e Qwiic	
Size: User	Date: 2026-11-08
KiCad E.D.A. 10.0.3	**Rev: 16R-READABLE**
	Id: 4/12

16 RIGHE PROGRAMMABILI | VCC3V3 / GND / GPIO / HI-Z

A1 | pad 39 | ROW_I01

ADC S3 / ADC S31

A2 | pad 15 | ROW_I02

ADC S3 / DIG S31

A3 | pad 6 | ROW_I03

ADC S3 / DIG S31

A4 | pad 7 | ROW_I04

ADC S3 / DIG S31

A5 | pad 4 | ROW_I05

ADC S3 / DIG S31

A6 | pad 26 | ROW_I06

DIG S3 / ADC S31

A7 | pad 29 | ROW_I07

DIG S3 / ADC S31

A8 | pad 30 | ROW_I08

DIG S3 / ADC S31

B1 | pad 38 | ROW_R01

ADC S3 / ADC S31

B2 | pad 12 | ROW_R02

ADC S3 / DIG S31

B3 | pad 17 | ROW_R03

ADC S3 / DIG S31

B4 | pad 18 | ROW_R04

ADC S3 / DIG S31

B5 | pad 31 | ROW_R05

DIG S3 / ADC S31

B6 | pad 32 | ROW_R06

DIG S3 / ADC S31

B7 | pad 33 | ROW_R07

DIG S3 / ADC S31

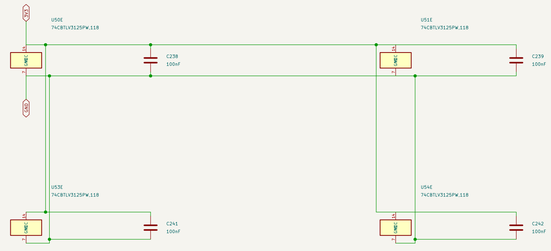
B8 | pad 34 | ROW_R08

DIG S3 / ADC S31

ADC DISTRIBUITI: 4 → S 53 / 4 S31: 8 → 4 S3 / 5 S31. Totale 9 / 9.
S31 → arrivi sul foglio principale U18…U. → contatti reali delle uscite socket.
J18…J31: le sette A…H sono gli 8 contatti della stessa connettore fisico.
Percorso continuo per ogni riga. U1 → isolatore → 4 contatti Hanca.
Le etichette sui fili mantengono soltanto le derivazioni verso mezzanine e controlli.

ALIMENTAZIONE DEI4ISOLATORI

Ogni integrato hancanali. UnitaE=alimentazione; bypass100nF.



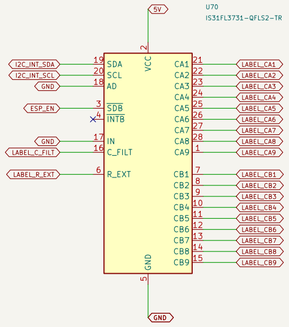
Sheet: /05 – Sedici righe programmabili/
File: ports16.kicad_sch

Title: 05 – Sedici righe programmabili

Size: A1 | Date: 2026-11-07 | **Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3 | Id: 5/12

DRIVER5V | I2C INTERNO3,3V |48LED

is31fl3731_0x74: SDB=ESP_EN3,3V. Nessun nuovo GPIO. R_EXT47kx1.38mAmedi/LED aPWM255.
CA16rossi, CB:16blu+16verdi. Avvio in software shutdown; azzerare enable/PWMprima di accendere.











A1 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A2 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A3 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A4 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A5 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A6 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A7 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

A8 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B1 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B2 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B3 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B4 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B5 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B6 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B7 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

B8 |3.3VROSSO /GNDBLU /MCU VERDE



LED45deg, scritte0deg, icona MCU=pinESP32,ADCsolo righe compatibili. HI-Z=tutto spento.

Sheet: /06 – Indicatori di configurazione retroilluminati/
File: rgb-ring.kicad_sch

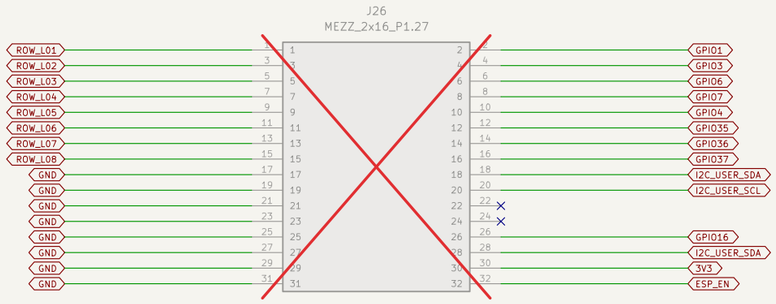
Title: 06 – Indicatori tre colori,5V,testi dritti

Size: A2	Date: 2026-11-08	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 6/12

J26 / BANCO A: FORI E GPIO SEPARATI

1..16: ROW = fori; GPIO = lato ESP, prima dello switch.

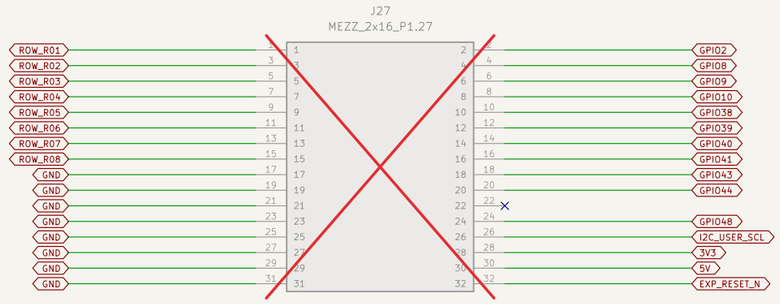
17..32: espansione, rail, masse e reset.



J27 / BANCO B: FORI E GPIO SEPARATI

1..16: ROW = fori; GPIO = lato ESP, prima dello switch.

17..32: espansione, rail, masse e reset.



BACKBONE PASSIVA: aggiunti solo J26/J27: nessun ADC/DAC, selettore, shunt o protezione extra.
Mezzanina assente: prese aperte. Presente: ingressi/uscite analogiche da progettare ad alta impedenza e sicuri anche senza alimentazione.
GPIO e ROW non vanno uniti sulla mezzanina: bypasserebbero gli switch e la separazione dal3V3/GND delle righe.
GPIO3.3V, non tolleranti5V. Righe limitate al dominio della backbone. I2C_USER condiviso con Qwiic/Hanxia, separato dagli expander; evitare conflitti di indirizzo.
ESP_EN: reset ESP, non abilitazione dedicata. EXP_RESET_N: uscitaU60, solo lettura; non comandarla dalla mezzanina.
Sincronizzazione e abilitazione: assegnabili nel firmware ai4GPIO ausiliari esposti anche suJ25 (pad9/25/36/37): GPIO11/12 riservati al bus utente, non a pin aggiuntivi dedicati.
La mezzanina di misura deve avere protezioni, disconnessione a riposo e sequenze sicure proprie; nessuna protezione nuova sulla base.
La corrente si misura instradando il circuito fra due righe distinte attraverso lo shunt della mezzanina; nessuno shunt sulla backbone.

Sheet: /07 – Mezzanina passiva/
File: mezzanine.kicad_sch

Title: 07 – Mezzanina passiva / tutti i nodi e GPIO

Size: A3	Date: 2026-11-08	**Rev: MEZZ-PASSIVE**
KiCad E.D.A. 10.0.3		Id: 7/12

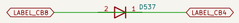
BLUE LOGO: see sheet11. Sixteen emitters, existing driver: no new GPIO or LED driver.

PLAY/PAUSE AND DEBUG

Driven by existing IS31FL3731; global touch inputs are on sheet10.



PLAY_PAUSE



DEBUG

LOCK / STOP / ROWS PAGE

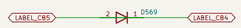
Driven by existing IS31FL3731; global touch inputs are on sheet10.



STOP



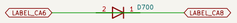
PAGE_ROWS



LOCK

CONFIG / SYSTEM / USER A / USER B PAGES

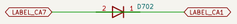
Driven by existing IS31FL3731; global touch inputs are on sheet10.



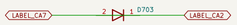
PAGE_CONFIG



PAGE_SYSTEM



PAGE_USER_A



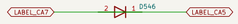
PAGE_USER_B

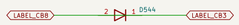
CONNECTION STATUS

Driven by existing IS31FL3731; global touch inputs are on sheet10.









CAN TERMINATION

Existing direct-drive LED retained.



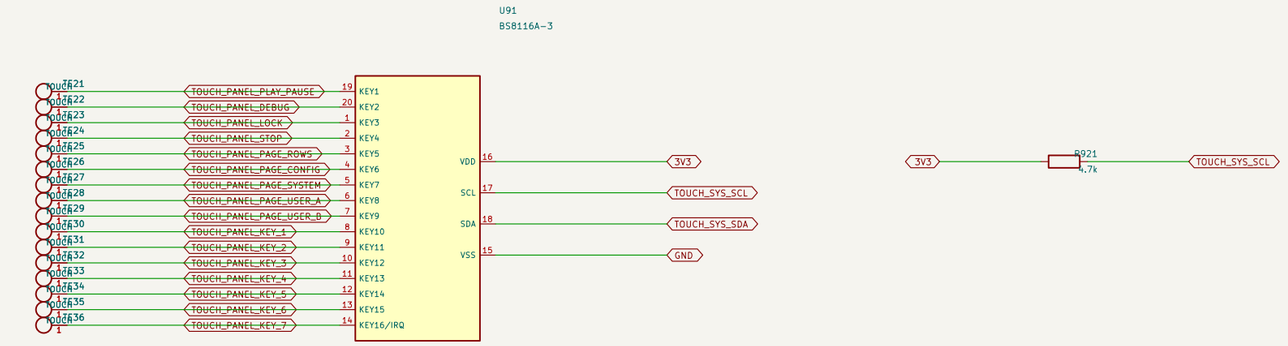


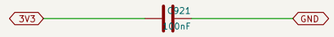
Sheet: /08 – Selezione, comandi e logo/
File: ui-indicators.kicad_sch

Title: 08 – LED status and touch command indicators

Size: A2	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 8/12

16 TOUCH PANEL: 7 shared keys + 5 page selectors + PLAY/PAUSE, DEBUG, LOCK, STOP.
STOP has priority even while LOCK/APPLY is active. Key16 is a touch input, IRQ disabled.
LOCK: only LOCK and STOP control LEDs enabled; row/status indications stay active.
Hold LOCK 2s to unlock. STOP preserves draft/slots; restart requires PLAY then YES.
Software I2C remains on module pads8/23; no user bus or row GPIO consumed.







Sheet: /10 – Global touch commands/
File: touch-global.kicad_sch

Title: 10 – Global and programmable user touch controls

Size: A3	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 10/12

SPAGHETTI LAB /16 BLUE REVERSE-MOUNT LEDS

D516..D519 retained: D720..D731 added. All16 rotated45degrees,3mm pitch.
Directed driver pairs are unique:77/144 channels populated,67free. No additional driver.

































Logo geometry traced from the user-supplied white-background original with preserved aspect ratio.
Light distribution through FR4, emitter-side routing, final LED current and optical uniformity require prototype qualification.

Sheet: /11 – Blue logo,16 LEDs/
File: logo16.kicad_sch

Title: 11 – Blue logo,16 emitters on existing LED driver

Size: A3	Date: 2026-11-07	**Rev: 16R-READABLE**
KiCad E.D.A. 10.0.3		Id: 11/12

Independent PWM channels. LOCK masks all control LEDs except LOCK and STOP.
Row selection/status remains visible. Optical windows require translucent overlay/light isolation.



KEY1_PAGE0



KEY1_PAGE1



KEY1_PAGE2



KEY1_PAGE3



KEY1_PAGE4



KEY2_PAGE0



KEY2_PAGE1



KEY2_PAGE2



KEY2_PAGE3



KEY2_PAGE4



KEY3_PAGE0



KEY3_PAGE1



KEY3_PAGE2



KEY3_PAGE3



KEY3_PAGE4



KEY4_PAGE0



KEY4_PAGE1



KEY4_PAGE2



KEY4_PAGE3



KEY4_PAGE4



KEY5_PAGE0



KEY5_PAGE1



KEY5_PAGE2



KEY5_PAGE3



KEY5_PAGE4



KEY6_PAGE0



KEY6_PAGE1



KEY6_PAGE2



KEY6_PAGE3



KEY6_PAGE4



KEY7_PAGE0



KEY7_PAGE1



KEY7_PAGE2



KEY7_PAGE3



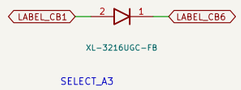
KEY7_PAGE4



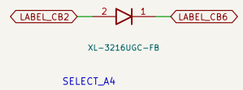
SELECT_A1



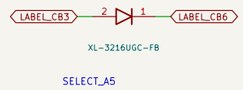
SELECT_A2



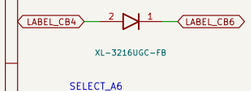
SELECT_A3



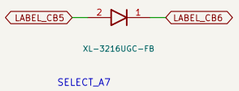
SELECT_A4



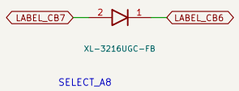
SELECT_A5



SELECT_A6



SELECT_A7



SELECT_A8



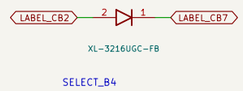
SELECT_B1



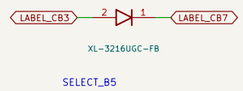
SELECT_B2



SELECT_B3



SELECT_B4



SELECT_B5



SELECT_B6



SELECT_B7



SELECT_B8

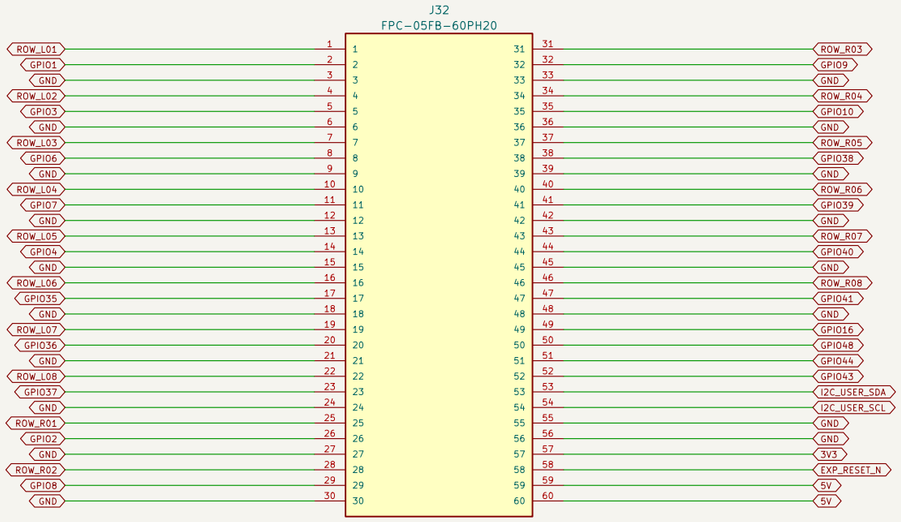


APPLY

Sheet: /12 – Panel LEDs/
File: panel-leds.kicad_sch

Title: 12 – Panel LEDs and row selection

Size: A3	Date:	**Rev: PANEL16-FFC60**
KiCad E.D.A. 10.0.3		Id: 12/12



FFC60 ADVANCED ACCESSORY — component side, cable through rounded slot
1..48: ROW / GPIO / GND triplets. Do not bridge ROW to GPIO on an accessory.
49..52: auxiliary GPIO aliases (UART0 may emit boot logs): 53/54: user I2C.
55/56 GND; 57 3V3; 58 EXP_RESET_N (LOW = outputs must be disabled); 59/60 5V.
STOP/boot: MCU holds ROW_ARM_N HIGH, EXP_RESET_N LOW. Accessory output stages
must default OFF and hardware-gate their OE with pin58; independent sources need local gating.
Power is shared, not isolated: no 5A claim through FFC. No hot-plug guarantee.
Use short cable; ADC accuracy and digital bandwidth depend on routing/cable/accessory.

Sheet: /13 — FFC60 accessory/		
File: expansion-ffc60.kicad_sch		
Title: 13 — FFC60 advanced accessory		
Size: A3	Date:	**Rev: PANEL16-FFC60**
KiCad E.D.A. 10.0.3		Id: 13/12